India's EV two-wheeler aftermarket: ₹869 Cr today, about ₹5,627 Cr by FY31

Market sizing of servicing and battery replacement, FY22–FY31 · bottom-up model with bear / base / bull scenarios

Executive summary: a small market today that grows about 6x in five years

₹869 Cr

Base-case aftermarket in FY26 (service and wear parts only; batteries not yet due)

₹5,627 Cr

FY31 base case; bear ₹3,622 Cr, bull ₹8,156 Cr

33%

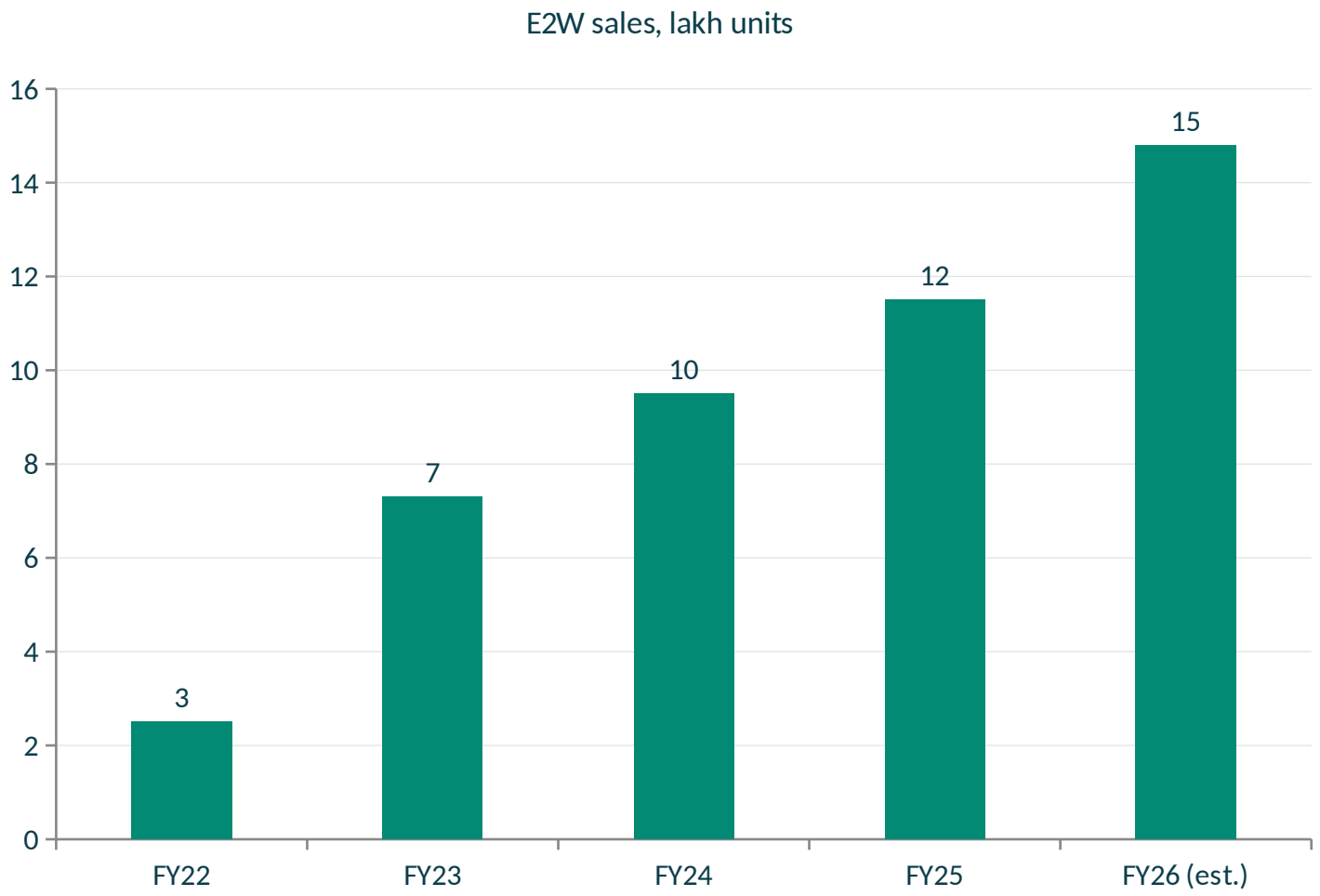
of the FY31 base market comes from battery replacements, a wave that starts in FY28

Growth comes from the fleet, not new sales alone: the fleet in use reaches about 16.3 million vehicles by FY31 in the base case, and every vehicle older than a year needs paid service.

The biggest swing factor is the battery wave: FY23–FY25 sales (2.8 million vehicles) reach replacement age from FY29, creating a lumpy, high-value repair stream.

Where to play: independent garages could address about ₹1,407 Cr by FY31; battery diagnostics, refurbishment and fleet contracts are the most attractive entry points.

Electric two-wheeler sales have grown about 6x since FY22



What the data says

- 2.5 lakh units in FY22 → 11.5 lakh in FY25 (Vahan-based government data).
- FY26 estimated at about 14.8 lakh: 2.55 million EV sales × ~58% two-wheeler share.
- Still only 6.3% of all two-wheeler sales in CY2025, so most growth is ahead.
- Growth slowed to +11% in CY2025: the base case assumes 20% a year, the bear case 10%.

Approach: build the market from the fleet up, then cross-check spend per vehicle



Sales by year

Vahan-based E2W sales FY22–FY26; FY27–FY31 grown by scenario

Fleet in use

Each year's fleet = last year's × (1 − 3% scrappage) + new sales

Service market

Vehicles past their free first year × annual service and wear-parts spend

Battery wave

Sales from 6 years earlier × share replacing × falling pack price

Cross-check

₹2,000–5,000 a year per e-scooter, about half a petrol scooter's cost

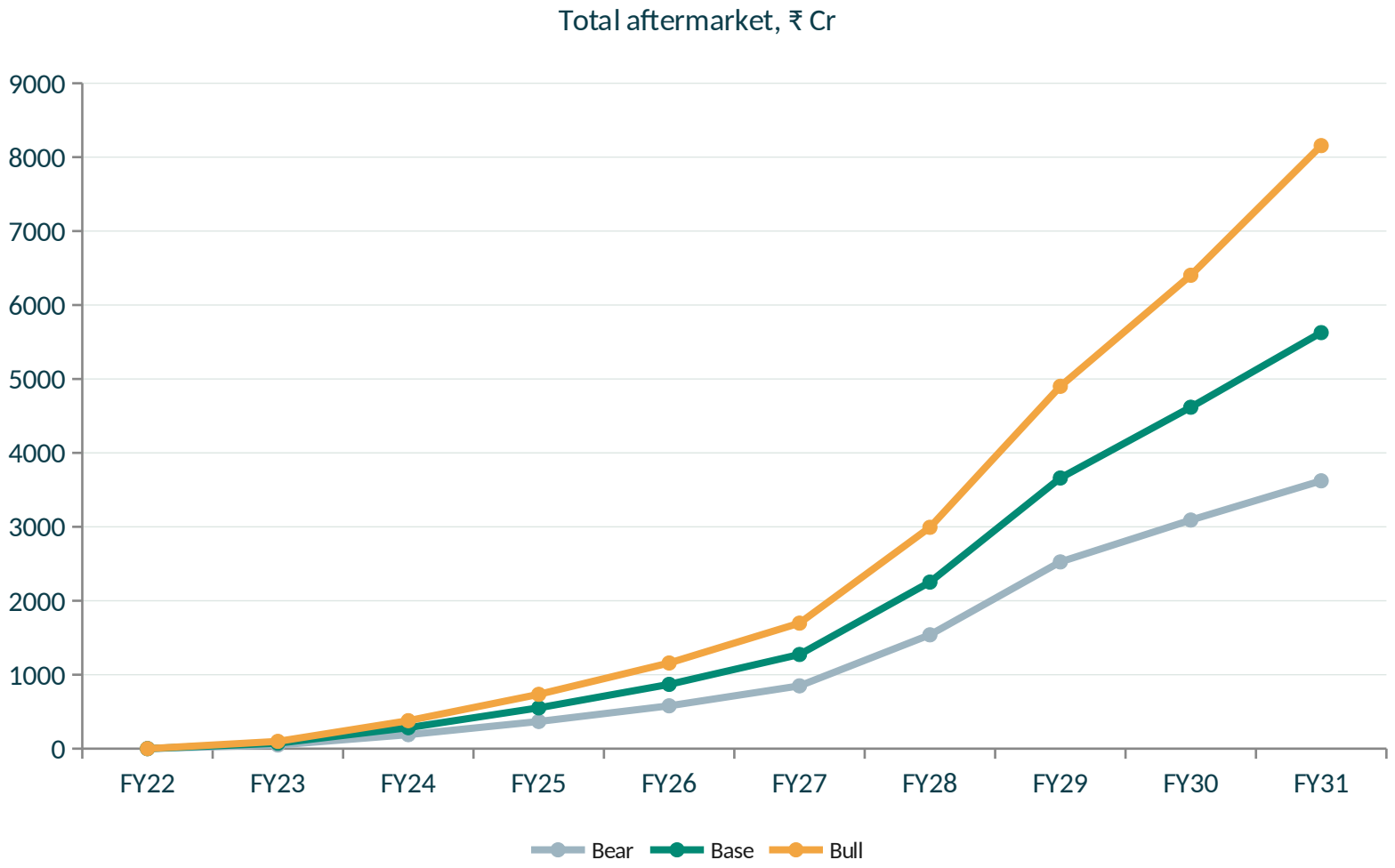
Why bottom-up: no published estimate exists for EV-specific aftermarket spend, and the EV fleet is young, so a top-down share of the petrol two-wheeler aftermarket would overstate it. Every input is in one Excel sheet, colour-coded as sourced or assumed.

Three scenarios bracket the uncertain inputs

Assumption	Bear	Base	Bull	Basis
Annual E2W sales growth, FY27–31	10%	20%	30%	CY25 +11%; FY25 +21%
Service + wear-parts spend per vehicle	₹2,000	₹3,000	₹4,000	Reported ₹2,000–5,000 a year
Share replacing battery at age 6	40%	50%	60%	Assumption; warranties 3–5 years
Scrappage per year	3%	3%	3%	Assumption
Battery pack price (FY26), yearly fall	₹55,000, −7%	₹55,000, −7%	₹55,000, −7%	Reported ₹55,000–60,000
Independent-garage share of spend	25%	25%	25%	Assumption

Biggest uncertainties to test first: spend per vehicle (no published average) and how many owners replace a battery rather than scrap the scooter.

The market reaches ₹3,622 Cr–₹8,156 Cr by FY31 across scenarios



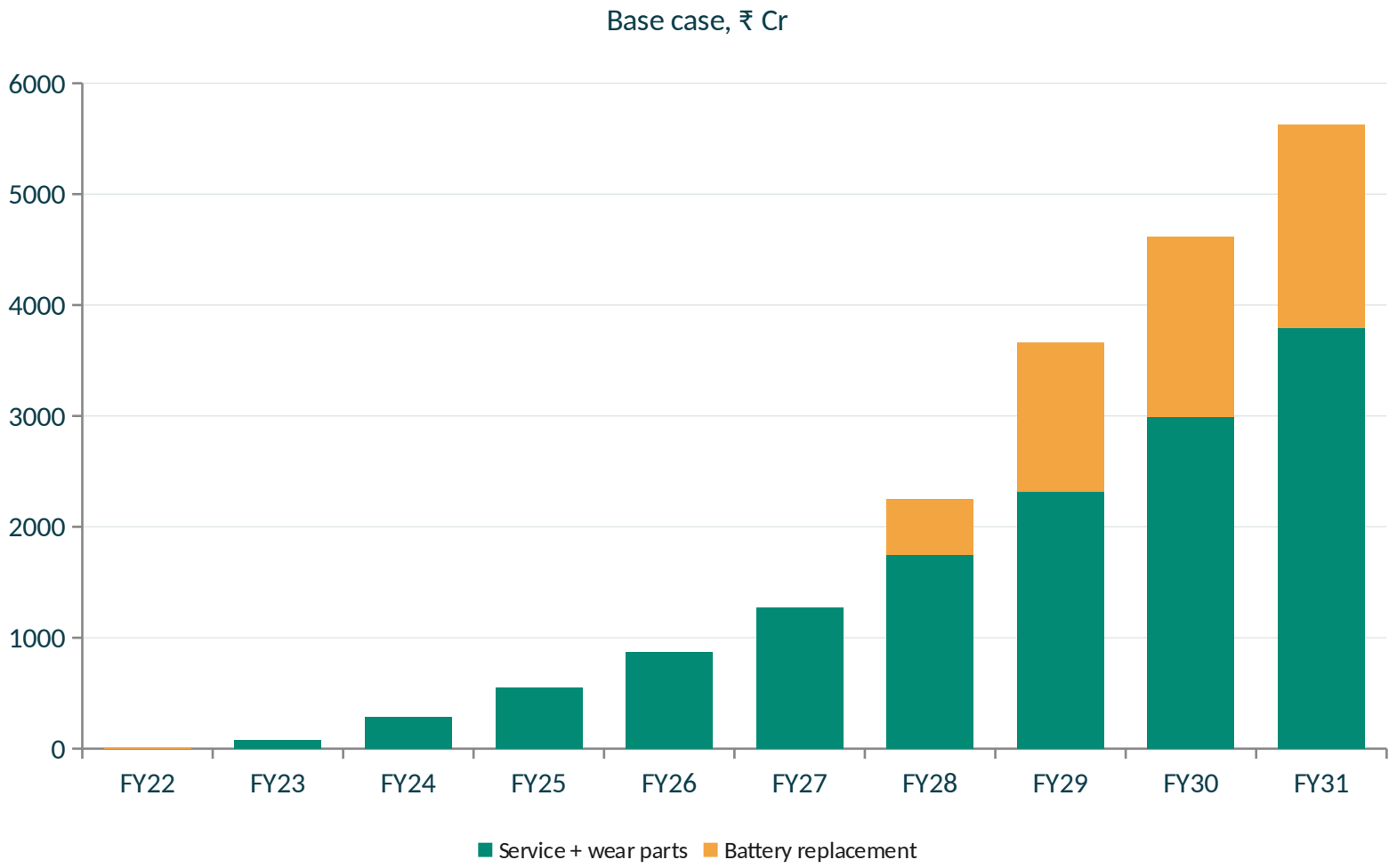
45% a year

Base-case growth FY26–FY31: fleet growth plus the battery wave

16.3 million

E2Ws in use by FY31 (base), up from about 4.4 million in FY26

From FY28, battery replacements become a second, lumpier revenue stream



Why the jump

- FY22's 2.5 lakh scooters reach age 6 in FY28; FY23's 7.3 lakh arrive in FY29.
- Each replacement is worth about ₹38,000–48,000 over FY28–FY31, versus ₹3,000 a year for routine service.
- Batteries are 33% of the FY31 base market, so battery health checks and refurbishment matter.

Spend per vehicle moves the answer more than sales growth does

Spend ↓ / growth →	10%	15%	20%	25%	30%
₹2,000	3,988	4,166	4,362	4,576	4,811
₹2,500	4,527	4,750	4,994	5,262	5,555
₹3,000	**5,066**	**5,334**	**5,627**	**5,948**	**6,300**
₹3,500	5,606	5,917	6,259	6,635	7,045
₹4,000	6,145	6,501	6,892	7,321	7,789

FY31 total aftermarket, ₹ Cr; other inputs at base case. Battery replacements in FY31 come from FY25 sales, so they don't change with future growth.

Reading the grid

- Moving spend from ₹2,000 to ₹4,000 adds about ₹2,200–3,000 Cr.
- Moving growth from 10% to 30% adds about ₹800–1,600 Cr, because FY31's fleet is mostly vehicles already sold.
- So the first piece of primary research should be what owners actually spend.

Where to play: independent service networks have a real, growing slice

₹1,407 Cr

FY31 spend addressable by independent garages (base, 25% share)

₹506 Cr

FY31 Karnataka market (base; ~9% of national EV sales)

~3,000

Rs per vehicle per year: price point for an annual maintenance plan

1. **Battery diagnostics and refurbishment:** Highest value per job; owners need advice on repair vs replace as warranties expire.
2. **Fleet contracts (delivery and rental):** High-mileage scooters service more often; one contract covers many vehicles.
3. **Multi-brand doorstep service:** OEM networks are likely thinner outside big cities (to verify); an on-site model avoids rent and serves several brands.

What would change the answer

Batteries last longer

If most packs outlive the scooter, the battery wave shrinks: the bear case already assumes only 40% replace.

OEMs lock in service

Software diagnostics and parts control can keep work inside OEM networks, cutting the independent share below 25%.

Sales growth stalls

Growth slowed to 11% in CY2025. At 10% a year (other inputs at base) FY31 is about 10% smaller; the full bear case is about a third smaller.

Data gaps

FY26 sales are an estimate from two sources, and there is no published average spend per EV.

Next steps: test the two inputs that matter most

1 **Owner survey (n ≈ 100)**
What do EV owners spend a year on service and parts, and where?

2 **10 garage and OEM-service interviews**
Job mix, prices, parts access, share of EV work today.

3 **Battery health data**
Warranty claims and pack degradation by age from 1–2 OEMs or fleet operators.

4 **Update the model**
Replace assumed inputs with survey medians; re-run the sensitivity grid.

Sources

- E2W sales FY22: Argus Media, citing Vahan (FY2021-22: 252,555 units).
- E2W sales FY23: PIB annexure, state and month-wise E-2W registrations (Vahan 4): 7,28,271.
- E2W sales FY24 and FY25: Ministry of Heavy Industries data via Times Drive (9,48,561 and 11,49,334).
- FY26 total EV sales: IESA India's Mobility Report FY2025-26 (2.55 million); E2W share ~58%: JMK Research.
- CY2025 E2W sales and penetration: EVreporter, Vahan dashboard (12,79,951; 6.3% of two-wheelers).
- Maintenance spend: MeraEV (2026), ₹2,000–5,000 a year for e-scooters vs ₹3,500–6,000 for petrol scooters.
- Battery replacement cost: e-vehicleinfo and BikeLeague (about ₹55,000–60,000).
- Karnataka share: IESA via IANS (about 9% of national EV sales in CY2025, all vehicle types).
- All other inputs are labelled assumptions in model/ev2w_service_market.xlsx (Inputs sheet).